Aneesh Grover

agrover@rockets.utoledo.edu LinkedIn GitHub Website

Experience

EventPin, Inc. **Sept 2026 – Jan 2027**
Software Engineering Intern *Holland, OH*

- Rebuilt the event-scraping pipeline (Scrapy, Python) across 4 sources, cutting one week's events needing manual review from 116 to 22 by automating rules the team previously applied by hand.
- Built an offline ZIP/state lookup (point-in-polygon on Census boundaries) that caught the upstream data source mislabeling **17%** of events, with no external API calls.

University of Toledo Learning Enhancement Center **Sept 2026 – Present**
Computer Science Tutor *Toledo, OH*

- Course tutor for Non-Linear Data Structures, Software Engineering, and DBMS.

Projects

folio.mcp - Personal MCP Server TypeScript, Cloudflare Workers, Hono, MCP SDK

- Deployed a remote MCP server on Cloudflare Workers exposing **4 tools** covering GitHub activity, portfolio data, and an LLM-invocable contact tool; **live in production**.
- Configured a KV layer for GitHub response caching with TTLs and IP-based rate limiting on the contact tool to prevent abuse.
- Shipped a companion frontend at **mcp.aneeshgrover.me** with dynamic tool discovery, backed by tests covering cache revalidation and rate-limit enforcement.

FocusBoard - Whiteboard Task Capture System Kotlin, FastAPI, CRAFT, TrOCR, Phi-3 Mini, TFLite

- Developed the Android app in Kotlin with ML Kit document scanning and an experimental on-device **YOLO/TFLite** whiteboard detector chosen for mobile size constraints.
- Engineered a **3-stage OCR pipeline** using CRAFT for line segmentation, TrOCR for transcription, and Phi-3 Mini for structured calendar JSON generation from handwritten input.
- Introduced per-device request queues to prevent race conditions under concurrent load.

Orbital - Cloud IDE Control Plane Go, Docker, JWT, REST APIs

- Built a Go control plane that provisions per-user VS Code workspaces on Docker and reconciles state every 30s so no container runs unaccounted for.
- Benchmarked workspace startup time and found it took **35x longer** the first time vs. later runs; traced the slowdown to a redundant image check on every request.

AcademicAlly React, Python, Pandas, BeautifulSoup

- Built a responsive timetable portal serving **10,000+ students** for instant class schedule lookup across campus; adopted by ACM Thapar Society.
- Wrote ETL pipelines processing **50,000+ Excel rows** with custom error handling, reducing parsing failures by **95%**.

Contributions & Involvement

Open Source Contributions Rust, TypeScript

- Designed and shipped `rustup doc --serve` to **rust-lang/rustup**, fixing sandboxed-browser `file://` doc-access failures; merged after maintainer review.
- Fixed a model-validation bypass in an open-source AI coding CLI (**merged PR**), closing a gap routing local model names to paid APIs.

Association for Computing Machinery (ACM), Thapar Chapter **Nov 2023 – May 2025**
Executive Board Member & Machine Learning Bootcamp Lead

- Organized Hacklipse 2024, ACM Thapar's hackathon, for **300+ participants**.
- Led ACM's Machine Learning Bootcamp curriculum, mentoring **100+ students** weekly.

Skills

Languages: Python, Go, Java, C++, JavaScript, TypeScript, SQL
Interfaces for Agents: Model Context Protocol (MCP), LLM tool integration, Cloudflare Workers, Hono, REST APIs
Infrastructure for Developers: distributed systems, WebSockets, JWT authentication, system design, Docker, CI/CD, Git
Data & Platforms: Supabase, PostgreSQL, Pandas

Education

University of Toledo **Toledo, OH**
Bachelor of Science in Computer Science and Engineering — GPA: 3.4/4.0 *Expected May 2028*

Achievements

- **Runner-up**, NVIDIA SatHack Open Innovation (**1500+** entrants) — Best Use Case, ACM Projectathon 2024.